Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Machine Learning and AI: Computer vision, Object detection and tracking, PyTorch, Scikit-learn, LangChain, RAG, LSTM, FAISS, Llama 3.2, Ollama, Pandas, NumPy, Flask

Cloud and Tools: AWS Lambda, LLM evaluation, Content understanding, Amazon S3, Prompt engineering, Amazon EC2, TensorFlow, Docker, Git, Vector embeddings, Hyperparameter tuning, Multimodal modeling

Data Engineering: Apache Kafka

Languages: SQL, Python

Projects

Stream Content Understanding Prototype *PyTorch, LangChain, FAISS, Llama 3.2*

- Build a prototype that classifies stream titles, chat text and channel metadata into content and brand-safety categories using open LLMs through LangChain
- Index stream text and metadata in a FAISS vector store to support semantic retrieval of similar content for labeling and review
- Compare LLM category outputs against human-labeled reference sets to measure agreement and guide prompt and model selection

MiGa - Multi-Chicken Gait Assessment (AICV Lab) *YOLO11n-Pose, OC-SORT, LSTM, PyTorch*

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements